
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

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Abstract

1 Learned forecasters increasingly stand in for scientific models that are themselves
2 imperfect, and a forecast score alone cannot say whether the surrogate succeeds for
3 the right reasons. We test five conditions that connect a score to a scientific claim
4 (held-out generalization, complete failure accounting, matched intervention, struc-
5 tural identifiability, and cross-system stability) on k -pendulum ($k \in \{1, 2, 3\}$) and
6 Lorenz-63 forecasts from language models, a time-series foundation model, learned
7 dynamics, sparse identification, and classical integrators. Every test changes a
8 substantive conclusion the unaudited comparison supports, from a fourfold in-
9 sample/out-of-sample gap, to exact symmetries that make the apparent parameter
10 recovery impossible, to a second system that reverses the model ordering. A cor-
11 rect leaderboard can still support incorrect scientific conclusions. We release the
12 prompts, outputs, checkpoints, seeds, and compute records needed to repeat the
13 audit.

14 1 Introduction

15 Scientific claims require more than low forecast error. When a learned forecaster stands in for an
16 imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched
17 inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of
18 the surrogate. Given state s_t , the task is to predict s_{t+T} ; we study these failure modes directly on
19 the k -link pendulum (integrable at $k=1$, chaotic for $k \geq 2$, high-precision ground truth, controllable
20 physical metadata) and Lorenz-63 (a different observation model and hypothesis class).

21 Our contribution is a claim audit for dynamical-system forecasts: held-out generalization, complete
22 failure accounting, matched disclosure on identical trajectories, identifiability checked before scoring
23 recovery, varied training budgets, and transfer to a second system. Table 1 states the conclusion before
24 and after each test; all five alter or reject a conclusion the unaudited comparison supports. Three
25 are the familiar concerns of imperfect models applied to the evaluation itself: misspecification (a
26 Hamiltonian prior on damped dynamics), identifiability (exact parameter degeneracies), and transfer
27 (a hypothesis class containing one system and not the other).

28 Appendix D situates the audited model families and the identifiability and selective-prediction
29 background.

30 2 Benchmark Design

31 2.1 Task and systems

32 Given the state $s_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to
33 predict s_T at horizon T . The pendulum varies the number of coupled links under a high-precision

Table 1: Conclusions before and after the claim audit.

Test	Unaudited conclusion	Audited conclusion
Generalization	Learned dynamics forecast accurately.	Error rises about fourfold; simple baselines perform better.
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Disclosure	Withholding constants improves forecasts.	Withholding constants increases pooled error by 0.069 rad.
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

numerical solution, and its constants can be hidden without changing the trajectory, which permits the matched test in §4.2; the smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system. Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4, the coarser-step RK4 baseline reported later is a numerical reference by construction, not an independent competitor.

2.2 Regimes and grid

Three regimes test generalization to non-standard constants (Table 8); in `changed_hidden` the constants (g, L, m , damping) are not disclosed to the model. An evaluation cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords, image, image+coords}\}$ and prompting $\{\text{direct, chain-of-thought}\}$ (both LLM-only). Non-LLM models use the coordinate/direct setting.

Out-of-sample protocol. Learned models are evaluated off their training data: a shared held-out set (a separate seed) of 180 initial conditions (60 per regime, $k \in \{1, 2, 3\}$) is scored identically for every family. The cross-family comparison uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated ($180 \times 2 = 360$ cells per model); we report no in-sample numbers and quantify the in-sample effect directly (§4.1).

2.3 Inputs and grouping

Table 3 (Appendix B) states what each family receives. Chronos is evaluated in its standard form, a history-conditioned forecaster with a 10 s pre- $t=0$ context window; the LLMs and learned dynamics models are state-conditioned from $t=0$. Learned hidden-regime cells are equalized by withholding hidden constants and using normal-regime priors. A separate asymmetry runs the other way: the LLMs are large proprietary models, the learned models small MLPs (Appendix B); larger physics-informed networks are outside our scope. Known-physics methods (RK4, Euler, symplectic) are numerical references, separated from black-box methods (LLMs, Chronos, learned dynamics).

2.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories.

Reliability-adjusted error. A cell is answered only if the model returns a parseable, finite prediction. Because scoring answered cells alone would reward abstention on hard cells, unanswered cells receive the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess); answering is part of the score.

Verification record. Each cell retains its identifiers, prompt, raw response, parser outcome, prediction, token counts, and score. Local methods are reproducible from the released checkpoints and seeds; hosted deployments can change or be withdrawn, so for them the released per-cell responses are the record, keeping the API evaluation auditable after the deployments it measured are gone.

71 **Disclosure gap.** We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate a penalty
 72 for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty,
 73 §4.2 also toggles disclosure on identical trajectories.

74 3 Methods

75 **LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through
 76 an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured
 77 prompt and JSON answer schema (Appendix A); each cell is one temperature-0 call with uniform
 78 token and timeout budgets, no best-of- N .

79 **Time-series foundation models.** We run Chronos [Ansari et al., 2024] locally in its standard
 80 history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels forecasts to the
 81 target horizon. The comparison uses `chronos-2` in univariate and multivariate modes; the paired
 82 mode comparison uses 20 trajectories per (k, regime) cell.

83 **Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and
 84 integrate at inference with `scipy`, comparing derivative matching, rollout, and mixed- k rollout
 85 objectives (Appendix B). The primary models train on nine trajectories, one per (k, regime) cell;
 86 a nested budget study trains fresh checkpoints at $\{1, 5, 20\}$ trajectories per eligible cell with three
 87 initializations (HNN: undamped cells only, since damping violates Hamiltonian conservation).

88 **Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations
 89 of motion; their error is from discretization alone.

90 **Baselines.** Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity
 91 ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast built from the
 92 same data as the learned models. On the balanced two-seed subset we also fit SINDy (degree-two
 93 library, STLSQ) and NVAR (two delays, degree-two features, ridge); divergent rollouts are retained
 94 as failures at $\pi/2$.

95 **Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence,
 96 SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on five
 97 five-trajectory held-out seeds, reporting training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s
 98 (configuration in Appendix C). HNN and LNN are omitted because Lorenz–63 is dissipative and
 99 non-Hamiltonian.

100 4 Results

101 4.1 Counting failed responses

102 The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model is
 103 evaluated on that grid, and the Answered column in Table 2 gives the fraction of cells with valid
 104 predictions: `grok` answers 82% and `l1nn` 94%, their ranks fall when missing cells receive $\pi/2$, and an
 105 answered-cells-only mean would give both a lower reported error. Intervals are 95% paired bootstrap
 106 over trajectories; LLM intervals exclude per-call sampling variation.

107 Table 2 ranks the models under the access conditions of §2.3; the ranking is conditional on those
 108 inputs, this training budget, and this system (§4.4 tests transfer). The known-equation integrators
 109 sit at the top as references, and every data-driven method is well behind them, closer to the $\pi/2$
 110 baseline than to the integrators. The lowest-error black-box entry is the LLM `kimi` (0.538 rad, 99%
 111 answered), whose interval excludes the nearest baseline: a statement about the recorded June 2026
 112 deployment, not a model-family ordering, which the Lorenz test rejects.

113 Among the trained models, after equalizing the hidden-regime inputs, nearest-neighbor lookup (0.725)
 114 and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but
 115 HNN-rollout (0.972), with the closest intervals overlapping. The stable learned models score about
 116 0.26 rad on their training trajectories and 0.97–1.14 rad held-out, so in-sample evaluation understates
 117 error about fourfold.

Table 2: Out-of-sample cross-family comparison (360 cells, horizons $\{1, 10\}$ s): reliability-adjusted mean angle error (rad), unanswered cells at $\pi/2$, 95% bootstrap CI. RK4 generates the data (reference). †: new in this work; ‡: simple baselines.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi†	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni†	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor‡	0.725	[0.657, 0.797]	100%	baseline
7	linearized‡	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence‡	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed†	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed†	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity‡	1.319	[1.241, 1.396]	100%	baseline
18	l1n† (full-scale)	1.356	[1.282, 1.430]	94%	learned

Two further data-driven methods appear only on the balanced two-seed subset (Table 4): SINDy (0.883 rad there) falls between the Chronos family and the learned models, and NVAR (1.248) is near constant velocity.

Effect of missing predictions. grok answers 82% of the grid and l1n 94%. Their ranks fall when missing cells receive $\pi/2$. An answered-cells-only mean omits the cases on which these models failed and gives both a lower reported error.

4.2 Matched disclosure test

On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). Read directly, this looks favorable to withholding constants, but the regimes use different trajectories, and the hidden ones are slower and less divergent, so the comparison cannot speak to parameter recovery. When we instead toggle disclosure on identical trajectories (Table 7), the pooled effect reverses: hiding the constants increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with disclosure helping at $k=1$ and at the 1 s horizon and only a small reverse effect at $k=3$. The unadjusted hidden-regime advantage is thus a difficulty confound.

The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto (am, ad)$ and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. A single-state probe returned mostly Earth-standard defaults; we treat those as descriptive and make no recovery claim. The released check verifies both invariances to below 10^{-13} .

4.3 Replication across seeds and training budgets

We reran the local pendulum subset on an independent held-out seed (five trajectories per cell versus 20; Table 4). The selected ordering is unchanged: numerical references lowest, Chronos-2 before the learned models, HNN-rollout best among them. Two seeds do not estimate seed-level uncertainty. The LNN’s low primary-seed coverage was a hardware artifact: its per-step Hessian exhausted the GPU on long rollouts; a CPU rerun recovers only the dropped cells (66% $\rightarrow$ 94%; both seeds use it), remaining failures are genuine divergences at $\pi/2$, and the mean stays 1.366 rad.

145 **Training-budget check.** Across three initializations and both held-out seeds, Neural ODE error
 146 falls from 1.090 at one trajectory per cell to 0.861 at twenty, and normal-regime HNN error from
 147 1.269 to 0.981 (Table 9); the rows cover different regimes and are not an architecture comparison.
 148 The primary result is partly data-limited, but more data does not close the gap to known physics.

149 4.4 Cross-system test: Lorenz-63

150 We test whether the pendulum conclusions transfer to Lorenz-63, a chaotic system with a polynomial
 151 vector field. They do not. The identifiability verdict flips: on the pendulum the physical constants are
 152 not recoverable from an angle trajectory (§4.2), while on Lorenz, SINDy recovers the vector field to
 153 normalized MAE below 2×10^{-5} at every horizon and in every seed. The difference is the setting,
 154 not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is fully observed, whereas the
 155 pendulum’s absolute constants are unidentifiable from angles under any method. Identifiability is a
 156 property of the observation model and the hypothesis class, which is what the check exposes.

157 The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25
 158 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and
 159 0.277 ± 0.063 at horizons 0.1, 0.5, and 1.0 s (mean $\pm$ standard error over seeds), below both Chronos-
 160 2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate
 161 similar). NVAR drifts to 0.429 by 1.0 s, persistence trails (0.900), and divergent rollouts score $\pi/2$ as
 162 on the pendulum (Table 4). This reverses the pendulum, where trained models lost to trivial baselines:
 163 here the system-trained model beats the zero-shot foundation model. As on the pendulum the families
 164 see different inputs (SINDy the library, the Neural ODE target-system training, Chronos a history
 165 window), so this is not an architecture race; the cross-system result rejects the pendulum ranking as a
 166 general ordering of model families.

167 The reversal does not rest on a favorable seed: the Neural ODE is lower in all 15 seed-by-horizon
 168 cells with roughly four-standard-error separation, and the identifiability half is not a sampling
 169 claim at all, since the pendulum symmetries are exact and Lorenz-63 lies in SINDy’s library by
 170 construction. The test excludes the API LLM grid and the Hamiltonian and Lagrangian networks (a
 171 Hamiltonian surrogate is misspecified for the dissipative Lorenz flow). Those omissions limit what it
 172 can establish, not what it can refute: one system on which both verdicts invert is enough to withdraw
 173 the generalization, and the Lorenz ordering is not offered as the correct one.

174 4.5 Diagnostics

175 Appendix E reports five further checks. Every black-box family saturates near the chaotic noise floor
 176 by 1 s, so $\{1, 10\}$ s is where families remain separable; error rises monotonically with k , and the
 177 hidden regime is easier for most models, confirming the difficulty confound of §4.2; rollout loss is not
 178 uniformly better than derivative matching; on the undamped regime the lowest learned-model energy
 179 drift (0.78, E^* -normalized) far exceeds the symplectic integrator (0.066), and a symplectic rollout of
 180 the HNN leaves its drift nearly unchanged, placing the misspecification in the learned Hamiltonian;
 181 and Chronos-2 multivariate and univariate modes are indistinguishable on 720 paired samples.

182 5 Discussion and Conclusion

183 Performance depends on the system, observations, hypothesis class, data, and access; a scalar
 184 leaderboard does not encode those conditions. Correct scores can support wrong scientific conclusions:
 185 each of the five checks changes a claim the raw leaderboard supports, and forecasting benchmarks
 186 should report the checks, training budgets, and information access as part of the result. We chose
 187 systems whose ground truth is cheap so each check could be shown to change a conclusion; the
 188 checks are worth most where the reference model is itself imperfect and verification is expensive. We
 189 release the prompts, outputs, checkpoints, seeds, and compute records needed to reproduce the audit.

190 **Limitations.** The full LLM grid uses one held-out seed and the local replication a second, unequal
 191 one; the budget study uses three initializations. Inputs differ by method (Table 3). The Lorenz test
 192 has five trajectories per seed and excludes API LLMs, HNN, and LNN; SINDy’s result depends on
 193 the library containing the true degree. Predictions are endpoints, conservation tests cover only the
 194 undamped regime, and foundation-model training corpora are unobservable.

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A Prompt templates

The prompts (from `bench/prompts.py`) are reproduced below.

Table 3: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

238 System prompt (direct / no-CoT).

239 You are a careful physicist predicting the future state of a k -pendulum (a chain of
 240 k point masses on massless rods, hinged in a planar chain, swinging under gravity
 241 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
 242 no code fences. Schema:
 243 {"theta": [...k floats..], "omega": [...k floats..]}
 244 theta_i is in radians (link i measured from straight-down,
 245 positive counter-clockwise)
 246 and omega_i is angular velocity in rad/s.

247 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,
 248 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a
 249 single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}.”

250 User prompt (coordinate modality).

251 Number of links $k = \{k\}$
 252 Physical constants:
 253 g (gravity, m/s^2): $\{g\}$
 254 L (rod lengths, m): $\{L\}$
 255 m (point masses, kg): $\{m\}$
 256 damping (viscous coefficient on omega): $\{\text{damping}\}$
 257 Initial state at $t=0$:
 258 theta_0 (rad): $\{\text{theta0}\}$
 259 omega_0 (rad/s): $\{\text{omega0}\}$
 260 Predict the state at $t = \{T\}$ seconds.
 261 Return theta and omega as length- $\{k\}$ JSON arrays.

262 **Hidden-constants regime.** The constants block above is replaced by:

263 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
 264 masses, and damping are NOT disclosed. You must infer dynamics from the initial
 265 conditions alone.

266 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
 267 `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k theta
 268 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

269 B Training and compute

270 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a
 271 single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
 272 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
 273 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
 274 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
 275 trajectories, one for each (k , regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
 276 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
 277 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

278 **Training objectives.** Derivative matching minimizes MSE on finite-difference $\dot{s}$; rollout loss minimizes
 279 MSE on the state after k Euler steps with gradients through the unroll; mixed- k rollout trains on several window
 280 lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order

autodifferentiation, which Apple’s MPS backend does not run, so it is trained at full scale (67k samples, hidden 256) on a CUDA GPU. Chronos-2 resamples the context to up to 512 points and forecasts ≤ 64 steps; the earlier `chronos-t5-large` run was preliminary, and Moirai’s `uni2ts` could not be installed in our environment.

Reproducibility records. The repository holds the code, prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep. The two large sweeps, the training-budget and balanced-seed studies, are published as release assets. Pendulum training uses seed 42; the primary held-out seed is 20260620 (180 trajectories), and the selected local subset also uses seed 20260621 (45 trajectories). LLMs are sampled once at temperature 0. The reported hosted-model numbers recompute from the saved records; re-calling the API is a new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

C Seed and system replication

Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, training seed 42, and held-out seeds 20260620–20260624. Each held-out seed contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library and STLSQ; NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz–63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	seed 20260620	seed 20260621	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

D Related work

The audited models span learned dynamics [Chen et al., 2018, Greydanus et al., 2019, Cranmer et al., 2020] on chaotic testbeds [Gilpin, 2021], zero-shot LLM and foundation-model forecasting [Gruver et al., 2023, Zhang and Gilpin, 2025, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], LLM physical reasoning [Lewkowycz et al., 2022, Wei et al., 2022, Xu et al., 2025], and sparse identification [Brunton et al., 2016, Gauthier et al., 2021]. The identifiability check draws on structural identifiability under parameter symmetries [Borgqvist et al., 2024]. Unlike selective prediction, which reports risk together with coverage [Geifman and El-Yaniv, 2017], failed cells here stay in the main score because models are compared at fixed task coverage.

E Extended diagnostics

This appendix expands the diagnostics summarized in §4.5.

Behavior by horizon. Out-of-sample error by horizon appears in Table 12. At 1 s the ordering follows the main comparison: the integrators are near-exact, `kimi` (0.457) has the lowest black-box error, and the learned models are at 0.86–1.26. At 10 s all models are in the chaotic regime, and the ordering is stable: `kimi` (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.44. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is mid-pack here (0.951 / 1.120). Figure 1 shows the growth curves over the {0.01, 1, 10, 60} s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The {1, 10} s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 5 and 6 report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for `kimi`, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The `changed_hidden` regime has the lowest error for almost every model (for `kimi`, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19; the previous values favored models given the true constants. The low-gravity, short-rod constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§4.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.

Training-objective check. Table 10 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (`normal`) regime, where true energy is constant. Table 11 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so changing the integrator does not account for the HNN’s drift. At matched objective, HNN drift is comparable to Neural ODE drift; the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

Predictability horizon. Pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ from $t=0$; the contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. Table 13 sweeps τ .

F Breakdowns by link count and regime

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons {1, 10} s), by number of links (Table 5) and by regime (Table 6); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 2.

G Additional result tables

Full per-model values for the horizon breakdown (§4.5) and the predictability horizon (§4.2), together with the regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

Table 5: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

Table 6: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity/short-rod constants give slower, less-divergent motion. The unadjusted regime contrast in §4.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 7: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

Table 8: Physical regimes. changed_hidden withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

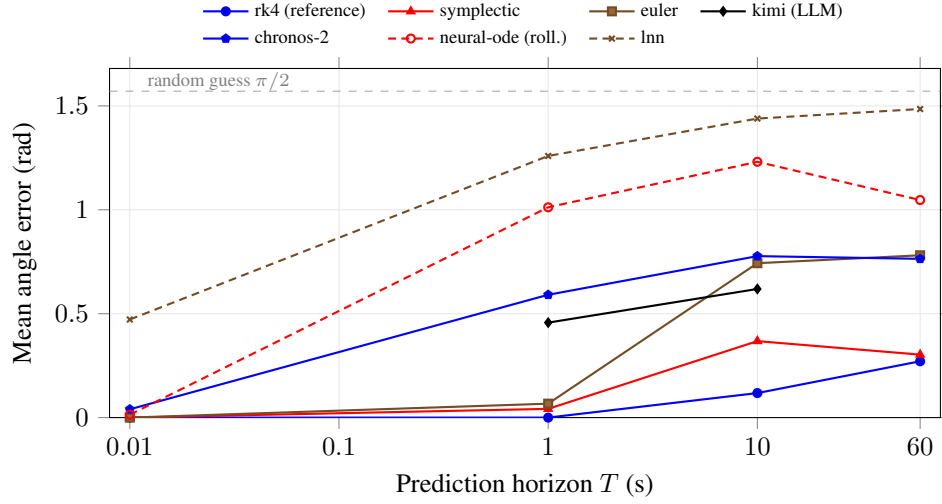


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 9: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 10: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 11: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 12: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimik2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.259	1.439

Table 13: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01